Student Performance Analysis

Data Science Research Test Dataset

1. Dataset Description

The Student Performance dataset represents a small educational analytics scenario. Each student record contains study hours per week, attendance percentage, assignment score, and final exam score. The final exam score can be treated as the primary outcome variable. The dataset can be used to demonstrate exploratory analysis, correlation analysis, regression, and the comparison of simple predictive models.

2. Variables

Student_ID is a unique identifier. Study_Hours represents approximate weekly study time. Attendance_Percentage represents the percentage of scheduled classes attended. Assignment_Score represents coursework performance as a percentage. Final_Exam_Score represents the final examination percentage. Student_ID should normally not be used as a predictive feature because it identifies a record rather than describing student performance.

3. Example Records

The following observations are illustrative research-test data. They are designed for retrieval and reasoning tests rather than as evidence about real students.

ID	Study Hours	Attendance %	Assignment %	Final Exam %
S001	5	82	74	76
S002	8	91	86	89
S003	3	70	65	68
S004	10	96	92	94
S005	6	85	78	81
S006	2	64	58	61
S007	7	88	83	85
S008	4	76	71	72
S009	9	93	89	91
S010	5	80	75	77
S011	11	97	95	96
S012	3	72	62	66
S013	6	87	80	82
S014	8	90	88	90
S015	4	79	68	71
S016	7	84	81	84
S017	2	67	55	60
S018	9	94	90	92

4. Analytical Interpretation

A researcher may investigate whether students with higher study hours tend to have higher final scores. Attendance and assignment score can be examined in the same way. A scatter plot can visualize numerical relationships, while correlation coefficients can summarize linear association. These analyses should be interpreted carefully because association does not establish a causal relationship.

5. Modeling Task

Final_Exam_Score can be selected as the regression target. Study_Hours, Attendance_Percentage, and Assignment_Score can be used as candidate features. A baseline linear regression model can be compared with a tree-based regression model. Both models should use the same training and test partitions for a fair comparison.

6. Evaluation Plan

For regression, MAE measures the average absolute prediction error, while RMSE gives greater weight to larger errors. R-squared describes the proportion of variance explained by the model under the chosen evaluation setup. A useful research table should report the model name, MAE, RMSE, R-squared, and prediction time.

7. Research Questions

Possible research questions include: Which variable has the strongest relationship with final exam score? Does adding attendance improve prediction compared with study hours alone? Which model provides the best balance between prediction error and computational cost? Are the conclusions stable when the train-test split changes?

8. Example Research Conclusion

A valid conclusion must be based on the measured results. If one model has a lower RMSE and MAE on the same test set, it can be described as having better predictive performance for that experiment. If another model is substantially faster while producing similar error, it may be preferable for applications where response time matters. The conclusion should state the experimental conditions and avoid claiming that the result applies universally.